

# ESP32-S3 Water Potability & Treatment Guidelines

Technical Knowledge Base & RAG Grounding Manual | Version 2.5 (Expanded Diagnostics)

**Operational Purpose:** Comprehensive specification document establishing parameter thresholds, deterministic potability evaluation logic, root-cause diagnostics, hardware telemetry protocols, and automated treatment pipeline architectures optimized for ESP32-S3 TFT displays.

## 1. Primary Parameter Thresholds & Operational Ranges

Water telemetry is continuously evaluated against three primary physical and chemical parameters: Potential of Hydrogen (pH), Total Dissolved Solids (TDS), and Electrical Conductivity (EC).

### 1.1 pH Classification Spectrum (1.0 to 14.0)

- **Potable (Ideal):** 6.5 to 8.5 (Balanced neutral state)
- **Acidic Range:** 1.0 to 6.49 (Corrosive risk)
- **Alkaline Range:** 8.51 to 14.0 (Scaling/caustic risk)

### 1.2 TDS Classification (mg/L or ppm)

- **Low / Flat:** 0.0-49.9 (Mineral deficient)
- **Ideal Quality:** 50.0-300.0 (Optimal balance)
- **Good Quality:** 300.1-500.0 (Acceptable tap)
- **Poor Quality:** 500.1-1000.0 (Mineral heavy)
- **Unacceptable:** > 1000.0 (Brackish/Saline)

### 1.3 Electrical Conductivity - EC (µS/cm)

Core conversion mathematical formula:  $EC = TDS / 0.65$

- **Ideal Range:** 0.0 to 400.0 µS/cm (Low mineral conductivity)
- **Acceptable Range:** 400.1 to 800.0 µS/cm (Standard municipal water)
- **High Range:** 800.1 to 1500.0 µS/cm (Elevated ionic concentration)
- **Severe Range:** > 1500.0 µS/cm (High salinity contamination)

## 2. Potability Evaluation Logic & Decision Matrix

**Deterministic Rule:** To classify as POTABLE, water telemetry must satisfy ALL conditions simultaneously: pH between 6.5 and 8.5, TDS between 50.0 and 500.0 mg/L, and EC between 0.0 and 800.0 µS/cm. Any single parameter violation immediately flags the water as NOT POTABLE.

### 3. Root Cause Analysis Matrix: pH Deviations (Acidic)

Granular diagnostic evaluation for acidic water anomalies and pipeline interventions.

| Violation Category      | Range      | Expanded Cause Analysis                                                                                                                                                                                                                                              | System Recommendation & Action                                                                                                                                                                                                                           |
|-------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Extremely Acidic</b> | 1.0 - 1.99 | <ul style="list-style-type: none"><li>The water is severely acidic, causing dangerous heavy metal leaching from pipes.</li><li>At this extreme level, the water is highly toxic and corrosive to nearly all standard plumbing materials.</li></ul>                   | <ul style="list-style-type: none"><li>Install an industrial-grade sodium hydroxide dosing system to aggressively neutralize the acid.</li><li>Shut down main water lines immediately and restrict usage until baseline pH levels are restored.</li></ul> |
| <b>Highly Acidic</b>    | 2.0 - 2.99 | <ul style="list-style-type: none"><li>The water exhibits high corrosivity, actively dissolving copper and lead fixtures.</li><li>Ingestion poses serious health risks due to the dissolved metallic contaminants entering the supply.</li></ul>                      | <ul style="list-style-type: none"><li>Install a high-capacity calcite and soda ash neutralizing tank to balance the pH.</li><li>Flush all plumbing lines thoroughly once the treatment system is active to clear residual metals.</li></ul>              |
| <b>Strongly Acidic</b>  | 3.0 - 3.99 | <ul style="list-style-type: none"><li>High acidity levels create a severe risk of structural damage to the water distribution pipes.</li><li>The water carries a distinctly metallic, sour taste and is completely unfit for human consumption.</li></ul>            | <ul style="list-style-type: none"><li>Deploy a soda ash injection pump paired with a Reverse Osmosis (RO) unit.</li><li>Monitor sensor telemetry daily to ensure the automated injection pump maintains a neutral pH output.</li></ul>                   |
| <b>Acidic</b>           | 4.0 - 4.99 | <ul style="list-style-type: none"><li>The water possesses a noticeable sour taste and accelerates gradual wear on metal pipes.</li><li>Continuous domestic use can eventually lead to pinhole leaks in household plumbing infrastructure.</li></ul>                  | <ul style="list-style-type: none"><li>Pass the water supply through a dual-tank limestone bed to elevate the alkalinity naturally.</li><li>Periodically backwash the limestone bed to prevent channeling and maintain treatment efficiency.</li></ul>    |
| <b>Mildly Acidic</b>    | 5.0 - 5.99 | <ul style="list-style-type: none"><li>Mild acidity introduces a metallic flavor and initiates slow, progressive plumbing corrosion.</li><li>Prolonged exposure can cause blue-green copper staining on sinks and bathroom fixtures over time.</li></ul>              | <ul style="list-style-type: none"><li>Install a standard calcite media filter to naturally raise the pH to safe, neutral levels.</li><li>Replenish the sacrificial calcite media annually, as it gradually dissolves into the treated water.</li></ul>   |
| <b>Slightly Acidic</b>  | 6.0 - 6.49 | <ul style="list-style-type: none"><li>The water sits just below the ideal limit, remaining mildly corrosive to sensitive metal plumbing.</li><li>While generally safer than lower pH levels, it still requires minor balancing to achieve full potability.</li></ul> | <ul style="list-style-type: none"><li>Install a simple inline remineralization cartridge to buffer and stabilize the pH.</li><li>Use this cartridge in conjunction with a standard carbon block to improve overall drinking taste.</li></ul>             |

### 3.1 Root Cause Analysis Matrix: pH Deviations (Alkaline)

Advanced diagnostic tables for extreme alkaline and scaling conditions.

| Violation Category         | Range        | Expanded Cause Analysis                                                                                                                                                                                                                                                 | System Recommendation & Action                                                                                                                                                                                                                                  |
|----------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Slightly Alkaline</b>   | 8.51 - 8.99  | <ul style="list-style-type: none"> <li>Elevated bicarbonate levels are present, which significantly reduces the effectiveness of chlorine.</li> <li>The water may feel slightly slippery to the touch and leave mild scale deposits on glass surfaces.</li> </ul>       | <ul style="list-style-type: none"> <li>Install a cation exchange resin filter to manage and reduce the excess alkalinity.</li> <li>Ensure regular sanitization of the water system, as the high pH lowers natural disinfection capabilities.</li> </ul>         |
| <b>Alkaline</b>            | 9.0 - 9.99   | <ul style="list-style-type: none"> <li>The water produces a bitter taste and actively causes hard mineral scale buildup in pipes.</li> <li>Thermal appliances like water heaters and kettles will suffer from heavily reduced heating efficiency.</li> </ul>            | <ul style="list-style-type: none"> <li>Install a weak acid softener system to continuously counteract and lower the high pH.</li> <li>Clean internal scale from existing appliances using a dedicated descaling solution once the water is treated.</li> </ul>  |
| <b>Strongly Alkaline</b>   | 10.0 - 10.99 | <ul style="list-style-type: none"> <li>The water causes heavy scaling blockages and presents a significant high pH chemical hazard.</li> <li>Prolonged skin contact during washing can cause severe dryness, irritation, and mild chemical burns.</li> </ul>            | <ul style="list-style-type: none"> <li>Install an automated HCl (hydrochloric acid) dosing pump to safely and predictably lower the pH.</li> <li>Implement strict automated pH monitoring to prevent any accidental acid overdosing in the pipeline.</li> </ul> |
| <b>Highly Alkaline</b>     | 11.0 - 11.99 | <ul style="list-style-type: none"> <li>The water is officially classified as a chemical hazard, carrying severe risks of tissue damage.</li> <li>It cannot be safely used for domestic washing, drinking, or standard agricultural processes.</li> </ul>                | <ul style="list-style-type: none"> <li>Install an aggressive acid dosing system followed immediately by a robust RO unit.</li> <li>Use extreme caution and wear appropriate personal protective equipment if handling raw water leaks.</li> </ul>               |
| <b>Extremely Alkaline</b>  | 12.0 - 12.99 | <ul style="list-style-type: none"> <li>The water is highly caustic, completely toxic to consume, and highly damaging to biological matter.</li> <li>Rapid pipe degradation resulting from caustic embrittlement is highly likely at this concentration.</li> </ul>      | <ul style="list-style-type: none"> <li>Implement a heavy, industrial-grade hydrochloric acid dosing loop for primary neutralization.</li> <li>Restrict all physical access to this water source until a full professional remediation is complete.</li> </ul>   |
| <b>Dangerously Caustic</b> | 13.0 - 13.99 | <ul style="list-style-type: none"> <li>The water is chemically equivalent to bleach, causing immediate skin irritation and severe burns.</li> <li>Vapors or incidental contact with this fluid pose extreme health, safety, and environmental emergencies.</li> </ul>   | <ul style="list-style-type: none"> <li>Shut the main supply valve immediately to halt the flow of the dangerously caustic water.</li> <li>Dose the standing water with neutralizing acid only under strict safety protocols and supervision.</li> </ul>         |
| <b>Max Alkalinity</b>      | 14.0         | <ul style="list-style-type: none"> <li>The water is at lye concentration, making it highly corrosive and instantly destructive to all tissue.</li> <li>This reading typically represents a critical system failure or a severe chemical contamination event.</li> </ul> | <ul style="list-style-type: none"> <li>Trigger an immediate system lockout and legally classify the water source as hazardous waste.</li> <li>Contact local environmental protection authorities for emergency spill containment and disposal.</li> </ul>       |

### 3.2 TDS & EC Evaluation Rules

| Violation Category       | Range                       | Expanded Cause Analysis                                                                                                                                                                                                                                               | System Recommendation & Action                                                                                                                                                                                                                                               |
|--------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Very Low Minerals</b> | TDS < 50<br>EC < 75         | <ul style="list-style-type: none"><li>The water lacks essential mineral ions, resulting in a distinctly flat and unappealing taste.</li><li>Because the water is overly pure, it becomes "hungry" and can aggressively leach metals from pipes.</li></ul>             | <ul style="list-style-type: none"><li>Install a dedicated calcium and magnesium remineralization cartridge into the flow path.</li><li>Ensure the remineralization cartridge is replaced every 6-12 months to maintain consistent output.</li></ul>                          |
| <b>High Minerals</b>     | TDS 500-1000<br>EC 800-1500 | <ul style="list-style-type: none"><li>The water is distinctly hard, leading to noticeable scale buildup and a potentially cloudy appearance.</li><li>This excess mineral density alters the taste of food and significantly reduces the lathering of soaps.</li></ul> | <ul style="list-style-type: none"><li>Install a 100GPD reverse osmosis (RO) membrane system to effectively strip out the excess minerals.</li><li>Pair the RO unit with a carbon pre-filter block to protect the sensitive membrane from premature fouling.</li></ul>        |
| <b>Severe TDS/EC</b>     | TDS > 1000<br>EC > 1500     | <ul style="list-style-type: none"><li>The water is classified as brackish, making it entirely unsafe and unsuitable for human consumption.</li><li>This high level of salinity will rapidly corrode and destroy standard household plumbing and appliances.</li></ul> | <ul style="list-style-type: none"><li>Install an industrial-grade water distiller or a sophisticated multi-pass RO system for heavy filtration.</li><li>Consider securing alternative freshwater sources if treating this highly brackish water proves too costly.</li></ul> |

### 4. Multi-Parameter Conflict Resolution

#### High TDS + Low pH (e.g., pH 5.4, TDS 850):

*Cause:* The water contains high dissolved solids that are acidic in nature.

*Action:* Install a reverse osmosis unit followed by an alkaline post-filter.

#### High TDS + High pH (e.g., pH 9.2, TDS 1100):

*Cause:* The water has high TDS composed of alkaline scaling minerals.

*Action:* Install a cation softener followed by a reverse osmosis unit.

#### Normal TDS + Extreme pH (e.g., pH 3.5, TDS 120):

*Cause:* The water suffers from severe chemical acid contamination.

*Action:* Install a sodium carbonate dosing pump to normalize the pH.

## 5. Granular TDS Quality & Taste Stratification

- **0.0-49.9 ppm:** Flat taste. *Action:* Add remineralizer.
- **50-150 ppm:** Ideal. *Action:* None.
- **150.1-300 ppm:** Tap quality. *Action:* None.
- **300.1-500 ppm:** Hard limit. *Action:* Carbon block.
- **500.1-1000 ppm:** Salty/scaling. *Action:* RO system.
- **1000.1-2000 ppm:** Brackish. *Action:* Dual-pass RO.
- **> 2000 ppm:** Saline (toxic). *Action:* Thermal distiller.

## 6. Temperature Compensation & 7. Sensor Diagnostics

**Baseline:** 25°C |  $EC_{25} = EC_{measure} / (1 + 0.02 \times (T_{measured} - 25))$  | **TDS** =  $EC_{25} \times 0.65$

- **Fault 1 (Dry Probe):** pH = 0.0-0.5 AND EC = 0.0  
*System Output:* "SENSOR FAULT: Probes are dry."
- **Fault 2 (Logic Conflict):** TDS > 0.0 BUT EC = 0.0  
*System Output:* "CALIBRATION FAULT: Check EC wiring."
- **Fault 3 (Out of Bounds):** pH < 0.0 OR pH > 14.0  
*System Output:* "CALIBRATION FAULT: Recalibrate pH."

## 8. Standardized Treatment Pipeline Architectures

|                                                                                                             |
|-------------------------------------------------------------------------------------------------------------|
| <p><b>Pipeline A (pH &lt; 6.5):</b><br/>Sediment Filter → Calcite Tank → Carbon Block → UV Lamp</p>         |
| <p><b>Pipeline B (TDS &gt; 500):</b><br/>Sediment Filter → Carbon Block → RO Membrane → Remineralizer</p>   |
| <p><b>Pipeline C (pH &gt; 8.5):</b><br/>Sediment Filter → Cation Softener → RO Membrane → Remineralizer</p> |